

Probing the User Representation with a Decorrelated Persona Pool

The “vulnerability axis” reconsidered — and a scientific direction

User-Axis study — expanded-design note

Abstract

Our first study found a single dominant *User Axis* on Llama-3.3-70B whose PC1 correlated with a held-out vulnerability tag at r up to 0.86. That pool, however, was built by spreading personas along a vulnerability/competence spectrum, so the tag space was effectively one-dimensional — a dominant PC1 was partly *imposed*, not *discovered*. This note rebuilds the dataset as a **decorrelated factorial** over ten user attributes, holds the probing pipeline fixed, and asks what the model’s user representation looks like when the attributes vary *independently*. The result overturns the headline: once vulnerability is decorrelated from competence and emotional load, it no longer drives PC1 ($\eta^2 = 0.005$); the dominant axes of user-representation variance are **topic, age, and competence**, while the psychological-state traits (vulnerability, trust, urgency, intent, style) are all weak ($\eta^2 \leq 0.10$). We argue that *dataset spread is the experiment* for this class of question, and lay out the direction it opens.

1 Motivation: from “is there a user axis?” to “what structures it?”

An instruction-tuned model silently infers who it is talking to and adapts. Our first study recovered that inference as a direction in activation space (the *User Axis* = PC1 of per-persona mean residual-stream vectors at layer 40), showed it correlated strongly with vulnerability, and showed it was behaviorally causal under steering. The natural next question is not *whether* such a direction exists but *what organizes it*: of all the ways users differ — expertise, age, topic, emotional state, intent, trust — which does the model actually encode as its primary axes?

That question cannot be answered by a pool whose attributes are entangled. If vulnerable users in the pool are also, by construction, less expert and more emotional, then a “vulnerability axis” and a “competence axis” and an “emotional-load axis” are the *same* direction, and PCA cannot tell them apart. **The dataset’s spread is the experiment.** So we change only the stimulus distribution — decorrelate the attributes — and keep every downstream measurement identical.

2 A decorrelated factorial user pool

2.1 Design

We define ten user-attribute *factors* (Table 1) and sample 300 personas (289 surviving generation) so that each factor’s levels are balanced and *crossed*: a balanced independent-shuffle followed by a short search that minimises the maximum pairwise Cramér’s V. This adds axes the original pool never varied on its own — intent, domain, communication style, urgency, and age as a factor rather than a correlate of vulnerability. All persona data (backstories, elicitations, tags) is generated by Claude Sonnet 4.5, one persona per design cell.

Table 1: The ten decorrelated user-attribute factors.

factor	levels
competence	novice · intermediate · expert
vulnerability	low · moderate · high
emotional load	calm · moderate · distressed
urgency	none · moderate · acute
trust	skeptical · neutral · deferential
intent	learn · decide · vent · create · verify
domain	health · finance · legal · coding · relationships · everyday
communication style	terse-technical · plain · verbose-casual
age	teen · young-adult · adult · older-adult
on behalf of	self · caregiver

2.2 Dataset spread: how decorrelated is it?

The design achieves **max pairwise Cramér’s V** = 0.10 (mean 0.02) across the ten factors — i.e. the attributes vary essentially independently (Fig. 1, left; contrast the ± 0.6 – 0.8 tag correlations of the lean-seeded pool, where expertise $\leftrightarrow$ vulnerability = -0.60 and vulnerability $\leftrightarrow$ emotional-load = $+0.77$). The decorrelation propagates to the observable judge-tags: the tag-space PCA spectrum *flattens* — PC1’s share falls from 0.60 to 0.50 and PC2 rises from 0.17 to 0.25 (Fig. 1, right). The residual vulnerability $\leftrightarrow$ emotional-load coupling that remains is a property of those two *scales* being near-synonymous to any annotator, not of the pool — which is exactly why we regress against the decorrelated *factors*, not the five legacy tags.

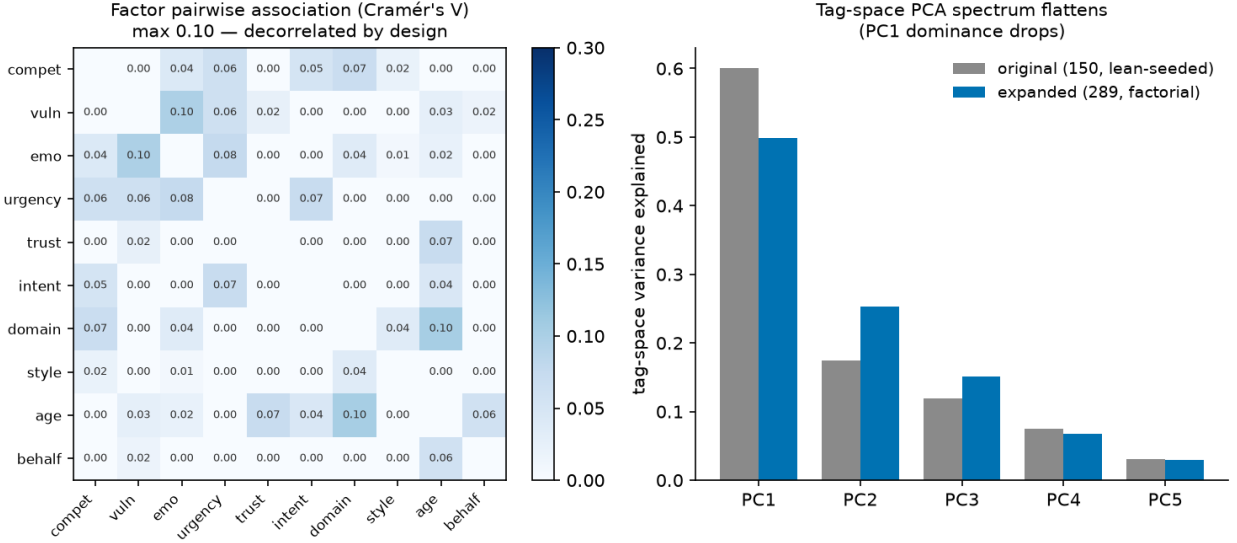


Figure 1: **Dataset spread.** *Left:* pairwise association (Cramér’s V) between the ten factors over 289 personas — all off-diagonal ≤ 0.10 . *Right:* tag-space PCA variance, original (150, lean-seeded) vs. expanded (289, factorial); the spectrum flattens, i.e. the pool is no longer effectively one-dimensional.

The probing pipeline is unchanged from the first study: the same dual elicitation (5 explicit system-prompt phrasings, 10 in-voice openers), the same 24 shared persona-agnostic probes, the same

`resp_mean/last_user` residual-stream readouts at all 80 layers, the same verbalized-inference filter (Stage D keeps 95.9% here, MAD 1.20, even cleaner than the original 94.0%), and PCA at $\ell = 40$. Holding the measurement fixed makes any change in the discovered axis structure attributable to the decorrelated design.

3 What the balanced pool reveals

We regress each principal component’s per-persona loading on each factor, reporting η^2 (fraction of the PC’s spread explained by that factor; one-way ANOVA, BH-FDR).

The vulnerability axis dissolves. PC1’s correlation with the vulnerability *tag* drops from 0.78 to 0.54 (`resp_mean`) and from 0.86 to 0.55 (`last_user`), and PC1’s variance share falls from 0.20 to 0.17. Decisively, PC1’s η^2 for the *independent* vulnerability factor is **0.005** — essentially zero. What PC1 actually tracks at $\ell = 40$ is **competence** ($\eta^2 = 0.17$), **emotional load** (0.13), and **topic** (0.13). In other words, the original “PC1 = vulnerability” finding was inflated by vulnerability being entangled with competence and emotional load; the pure, decorrelated vulnerability signal is not a dominant direction.

The dominant axes are topic, age, and competence. Across PC1–PC5 (Fig. 2), the strong movers are **domain** (η^2 up to 0.64 on the `last_user` readout), **age** (up to 0.45), and **competence** (up to 0.36). Crucially, domain dominates even on the *pre-response* `last_user` token — so it is the model organising its user representation by topic, not merely response content leaking into `resp_mean`.

The psychological-state traits are weakly represented. Vulnerability, trust, urgency, intent, communication style, and self-vs-caregiver are all weak ($\eta^2 \leq 0.10$) on every one of the top five components. Whatever the model encodes most strongly about its user, in a balanced pool, is *who they are* (topic, age, expertise) far more than *what state they are in*.

4 Comparison to the lean-seeded pool

Table 2 makes the reversal explicit. The two studies use the same model, layer, probes, and pipeline; they differ only in how the persona pool is spread. That single change moves PC1 from “the vulnerability axis” to “a competence/topic axis on which the independent vulnerability manipulation has no measurable effect.”

This is not a contradiction of the first study so much as a correction of its interpretation. The axis is real and reproducible; what was over-claimed was its *identity*. A direction that separates the personas we *built* to differ in vulnerability will of course correlate with vulnerability — but that is a statement about the pool, not about the model’s priorities.

5 This as a scientific direction

The broader lesson generalises beyond vulnerability: **when you read a latent attribute off a model by contrasting hand-authored stimuli, the stimulus distribution can manufac-**

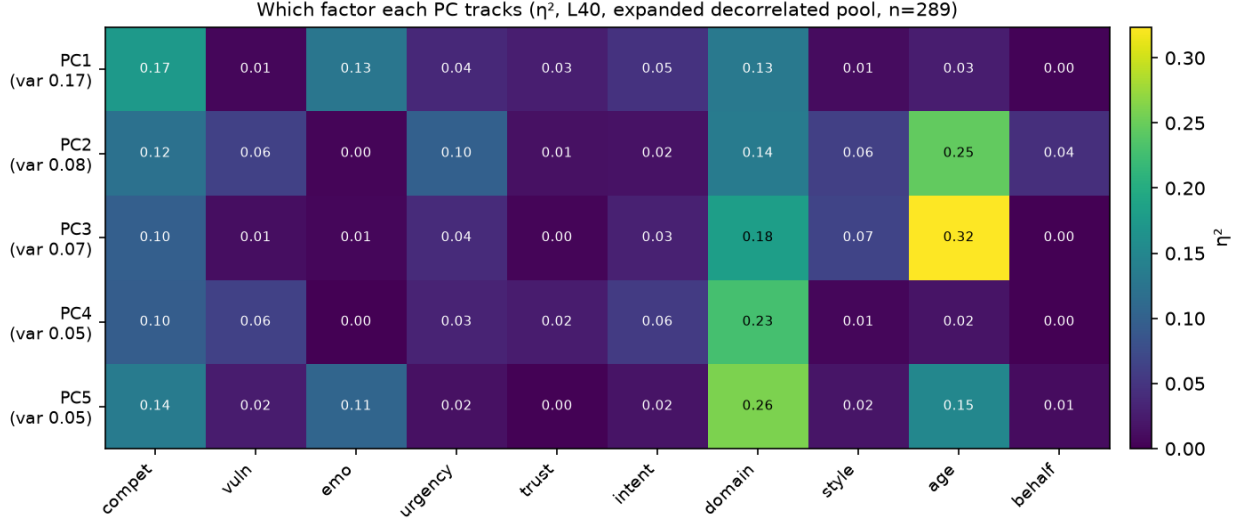


Figure 2: **Which factor each PC tracks** (η^2 , `resp_mean`, $\ell = 40$, $n = 289$). PC1 loads on competence/emotional-load/topic — *not* vulnerability ($\eta^2 = 0.01$). The bright columns are domain and age; the vulnerability, trust, urgency, intent, style, and on-behalf columns are dark throughout.

Table 2: Original (lean-seeded, $n = 150$) vs. expanded (decorrelated factorial, $n = 289$); `resp_mean` at $\ell = 40$.

	original	expanded
pool design	lean-seeded (22 archetypes)	decorrelated factorial (10 factors)
attribute entanglement	bundled ($ r $ up to 0.77)	decorrelated ($V \leq 0.10$)
tag-space PC1 share	0.60	0.50
PC1 activation variance	0.204	0.173
PC1 $\sim$ vulnerability tag	0.78	0.54
PC1 η^2 , vulnerability <i>factor</i>	— (confounded)	0.005
PC1 top factors	(vulnerability bundle)	competence, emo-load, topic
dominant axes overall	vulnerability	topic, age, competence

ture the very axis you report. De-correlating the design is the antidote, and it turns “does the model represent X ?” into a sharper, falsifiable question: *how much of the representation’s variance does an independently-manipulated X explain, holding everything else fixed?* Under that test, several attributes the field treats as central to “user modelling” (vulnerability, trust, intent) turn out to be weakly encoded, while topic and age dominate.

We see three concrete threads:

- **Disentangle topic from user-state.** Domain has the most levels and the largest η^2 ; partialling it out (or matching domain across the state factors) would sharpen how much genuine *state* structure remains. That the signal is strong even on the pre-response token suggests the model’s “user model” is substantially a “conversation-topic model.”
- **Blind, held-out axis naming on the balanced pool.** Re-running the unsupervised auto-labeling (a judge names the extreme poles of each PC, validated on held-out personas) on this pool would give an interpretation-free account of what topic/age/competence axes actually separate.

- **Causality per axis.** Steering along the new competence/age/topic directions — and testing whether steering moves the model’s *verbalized* user model, not just its register — would establish which of these representations are behaviourally load-bearing rather than merely present.

Caveats. One model, one layer family; **resp_mean** is response-content-laden (mitigated but not eliminated by the **last_user** check); and η^2 favours factors with more levels (domain has six). None of these overturn the central result: the dominant, decorrelated axes of the model’s user representation are demographic and topical, and the vulnerability axis was, to a substantial degree, a property of the pool we built.